B.TECH CSE · MAJOR PROJECT · OCTOBER 2026

Pramana

प्रमाण *proof; a means of valid knowledge*

AI-assisted legal research for Indian criminal law that verifies every citation it makes, across the IPC → BNS transition.

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OUTLINE

What we will cover.

01
The problem
what lawyers in India face

02
What exists
research, tools, the gap

03
Our solution
what Pramana does, and how

04
How it works
architecture, data, retrieval

05
Multi-agent pipeline
every agent, one by one

06
Technology
the stack and why

07
Demo
a real case, end to end

Criminal legal research in India is still done by hand.

WHO DOES IT

- Advocates, especially district-court practitioners
- Law students
- Legal-aid clinics

1
Bare act
find candidate sections

→

2
Commentary
read the ingredients

→

3
Search judgments
keyword search for precedent

→

4
Manual reading
hours per matter
[[VERIFY: time per matter]]

AND NOW

Two criminal codes since 1 July 2024
IPC and BNS in use side by side

AI tools invent citations
a wrong section is not a cosmetic error

Precedent is split across codes
new BNS sections have almost no case law yet

On 1 July 2024 India renumbered its criminal law.

INDIAN PENAL CODE, 1860 [2] offences before 1 July 2024	SECTION	BHARATIYA NYAYA SANHITA, 2023 [1] offences from 1 July 2024
Punishment for murder	302	Uttering words with deliberate intent to wound religious feelings
Punishment for murder by a life-convict	303	Theft
Cheating and dishonestly inducing delivery of property	420	*No such section. The BNS ends at 358.*

A section number means nothing without its code, and *the offence date decides the code.*

AI models cite law with confidence, including law that does not exist.

A LANGUAGE MODEL, ASKED DIRECTLY

Q What does BNS Section 303 cover?

A A public-panic offence: making or spreading a bomb threat.

× Wrong. BNS 303 is **Theft**.

20 *of* **68**

citations named sections not in the BNS when the model answered from memory.

15 development cases, retrieval withheld.

17–33%

of queries hallucinated by commercial legal AI research tools [6]. [[VERIFY: Magesh et al. figure]]

PROBLEM STATEMENT

Build a research assistant for Indian criminal law that identifies applicable sections and precedents across the IPC→BNS transition, and *never presents a non-existent citation*.

What exists, and what nobody combines.

RESEARCH [[VERIFY: paper details]]

[3] **RAG** · Lewis et al., NeurIPS 2020
grounds answers in retrieved text; no citation check

[4] **LeSICiN** · Paul et al., AAAI 2022
graph statute identification; supervised, IPC only

[5] **IL-TUR** · Joshi et al., ACL 2024
Indian legal benchmark; IPC-era labels

[6] **Legal AI hallucination** · Magesh et al., 2024
measures the failure, does not prevent it

[8] **BGE-M3** · Chen et al., 2024
general embedder; untested on Indian law

EXISTING TOOLS

	IPC/BNS BY DATE	VERIFIES CITATIONS	PRECEDENT ACROSS CODES	COUNTER-ARGUMENTS	
Indian Kanoon	✗	—	✗	✗	[[VERIFY]]
Manupatra / SCC Online	?	—	?	✗	[[VERIFY]]
jhana.ai	?	✓	?	?	[[VERIFY]]
NyayAssist	?	✓	?	?	[[VERIFY]]
Pramana	✓	✓	✓	✓	

— search engines and databases return real documents; they do not interpret facts.

THE GAP

No system combines **date-based code routing**, **graph-verified citations**, an **IPC→BNS precedent bridge**, **counter-arguments** and **outcome statistics**.

Facts and a date go in. A checked research memo comes out.

CASE FACTS · THE RUNNING EXAMPLE

The complainant parked his motorcycle outside his house at night. In the morning it was gone. CCTV shows the accused pushing it away down the lane and riding off. The accused had no permission to take it and did not return it.

OFFENCE DATE 15 Aug 2024

Pramana

CODE	BNS, because the offence is on or after 1 July 2024
CHARGE	BNS 303 Theft ✓ verified
PRECEDENTS	3 judgments, found through its IPC predecessor, IPC 379
DEFENCE	A counter-argument, its quote checked against the facts
OUTCOMES	How 158 appeals under IPC 379 were decided

The right code, every time
the offence date decides IPC or BNS

Every citation verified
a section is shown only if it exists in that code

Precedent across both codes
BNS charges reach IPC case law

Four steps: route, retrieve, verify, ground.



01

Route

The offence date picks the code: IPC or BNS. A fixed rule, no AI.

02

Retrieve

Find candidate sections from statute text and from similar past judgments.

03

Propose & verify

The AI proposes sections; the citation graph confirms each one exists.

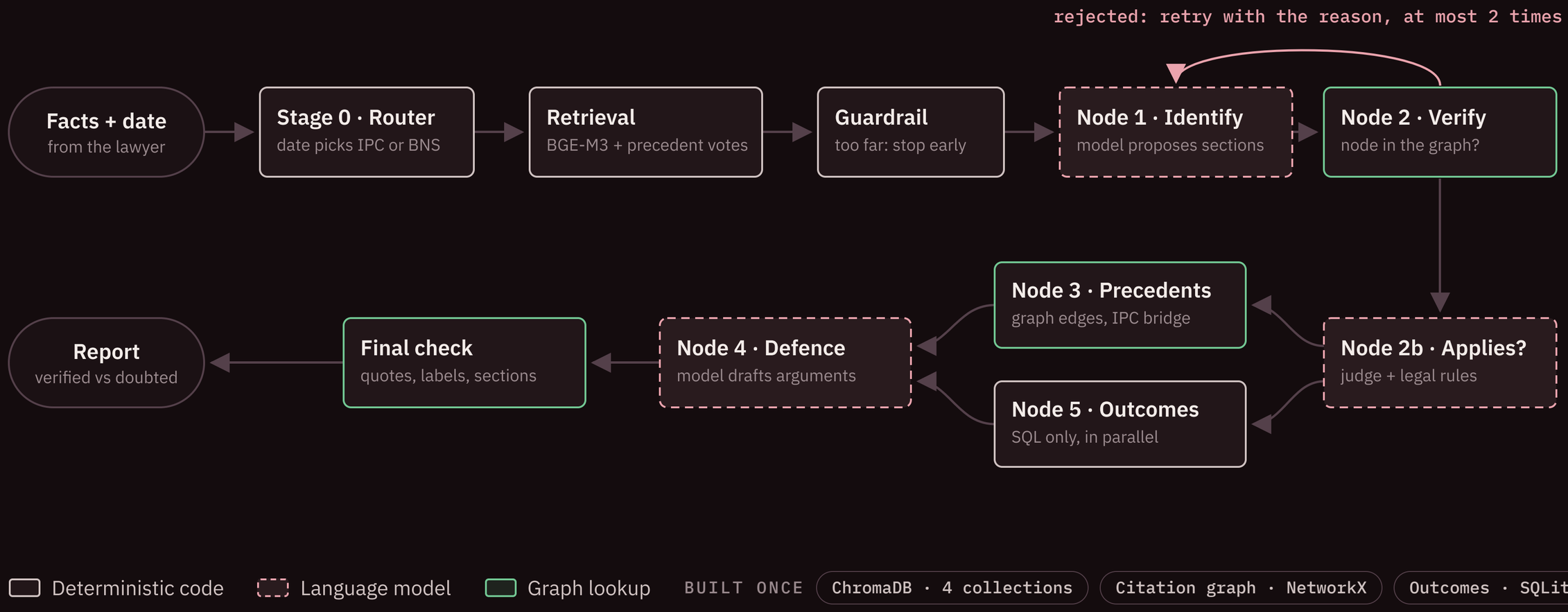
04

Ground & argue

Test each section on the facts, find precedents, draft counter-arguments.

The AI proposes. *The graph decides what is real.*

The model proposes. The graph decides.



From public sources to three stores.

SOURCES

IPC — 575
sections · structured JSON + official text [2]

BNS — 358
sections · parsed from the gazette PDF [1]

Judgments — 2,699
918 Supreme Court (2023–26) + 1,781 High Court [11]

IPC→BNS map — 575
rows, each reviewed against the gazette text

PROCESSING

1. **Parse** — PDF text with PyMuPDF; repair ligatures, strip page headers
2. **Chunk** — one section per chunk; judgments by paragraph
3. **Extract** — cited sections per chunk; CrPC and other Acts filtered out
4. **Validate** — every record against a fixed schema
5. **Embed** — BGE-M3 vectors, 1024 dimensions

STORES

ChromaDB · 4 collections
IPC sections · BNS sections · IPC-era judgments 83,918 · BNS-era 3,118
`87,036 passages`

Citation graph · NetworkX
933 section nodes · 2,699 judgments · 16,493 citation edges
`539 IPC→BNS bridges`

Outcomes · SQLite
appeal disposals per section

Retrieval finds the candidates; nothing is fine-tuned.

WHY RAG [3], NOT FINE-TUNING

Knowledge stays current
new judgments are added, not trained in

Every answer traces to a source
each candidate is a real indexed section

The model reads the real text
the section itself is in front of it when it proposes

THE MOTORCYCLE CASE, STEP BY STEP

1. **Date rule**
   15 Aug 2024 ≥ 1 Jul 2024 → **BNS**

2. **Embed the facts**
   BGE-M3 [8], 1024 dimensions

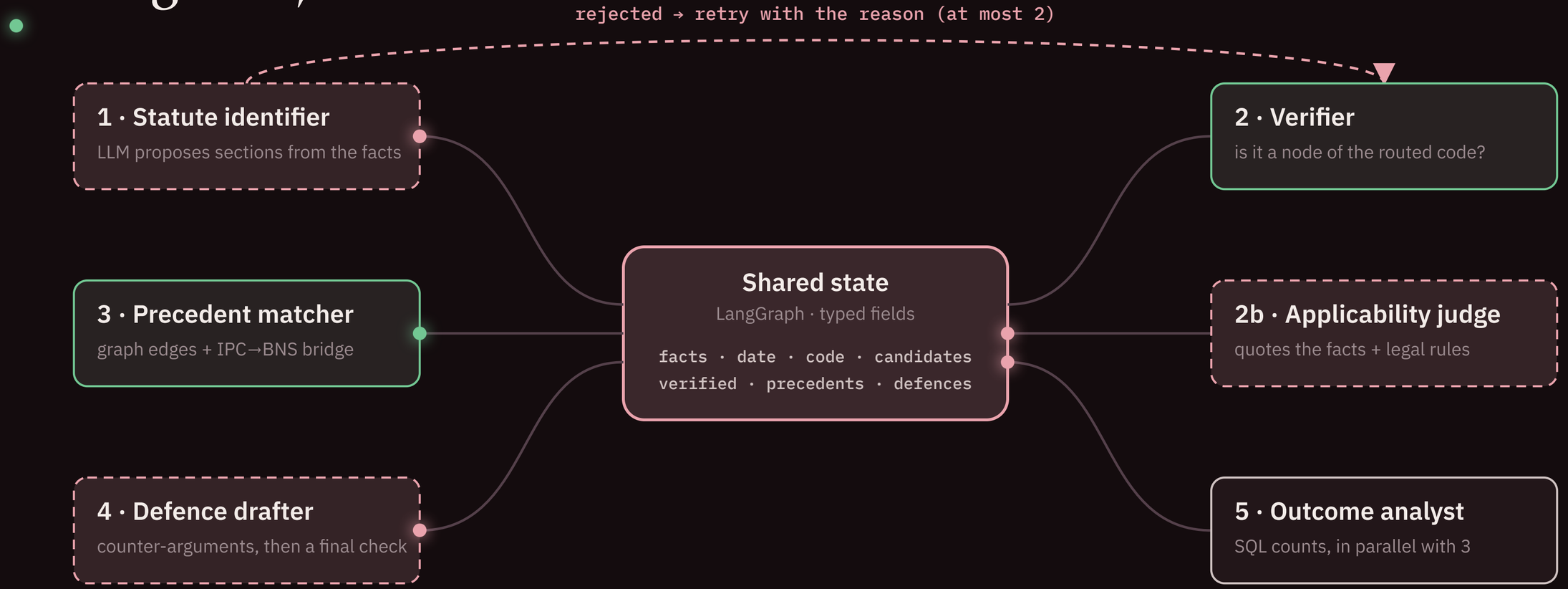
3. **Statute search + precedent votes**
   similar judgments vote for the sections they applied; fused by reciprocal rank fusion [9]

4. **14 candidates for the agents**

303(1)	Theft	0.0320 · T3 V2
317(5)	Stolen property	0.0310 · T6 V3
324(1)	Mischief	0.0265 · T5 V30

Guardrail: closest judgment 0.686, above the 0.576 threshold, so the agents are asked.

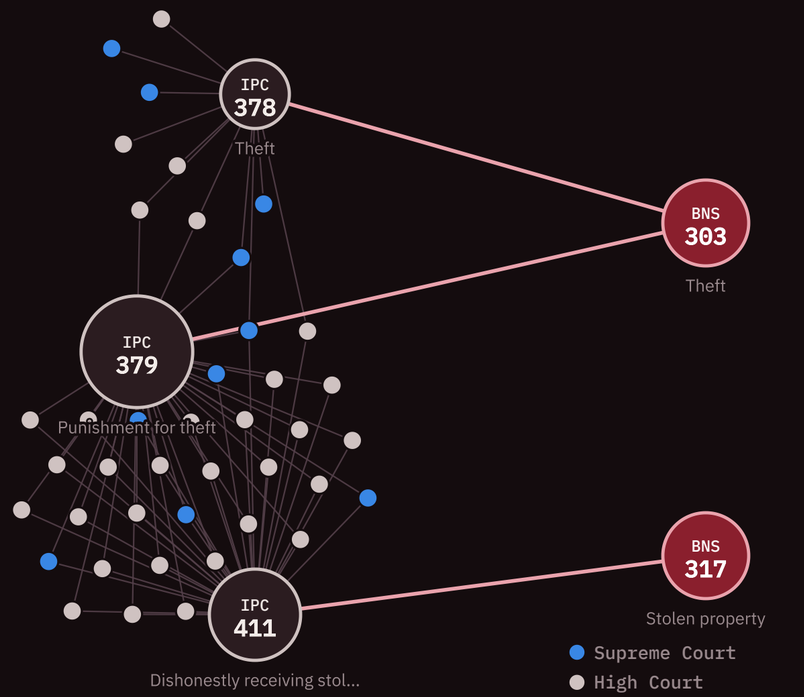
Six agents, one shared state.



Code · Language model · Graph lookup

Each agent reads and writes only its own fields of the state, so every step can be checked on its own.

Every citation is checked against the law itself.



933 sections · 2,699 judgments · 16,493 citation edges · 539 IPC→BNS bridges

THE RETRY LOOP · A REAL RUN WITH RETRIEVAL WITHHELD

ROUND 0 — identifier: 421

✕ Not a section of the BNS, which has sections 1 to 358.

ROUND 1 — identifier: 378

✕ An IPC number. Hint from the graph: its BNS counterpart is **BNS 303**.

ROUND 2 — identifier: 303

✓ **BNS 303 · Theft.** A node of the BNS. Verified.

20 / 20
real vs invented section numbers told apart

0
rejected citations ever reached a report

A real section can still be the wrong charge.

FACTS · KNIFE STABBING · 3 NOV 2024

After a quarrel over a parking space, the accused took a kitchen knife from his car and stabbed the complainant in the abdomen. Surgeons repaired a perforated intestine, and the medical report records the injury as dangerous to life.

LEGAL RULES
- death needed for homicide
- 5+ people for rioting
- lesser offence included
- weapon + injury grade

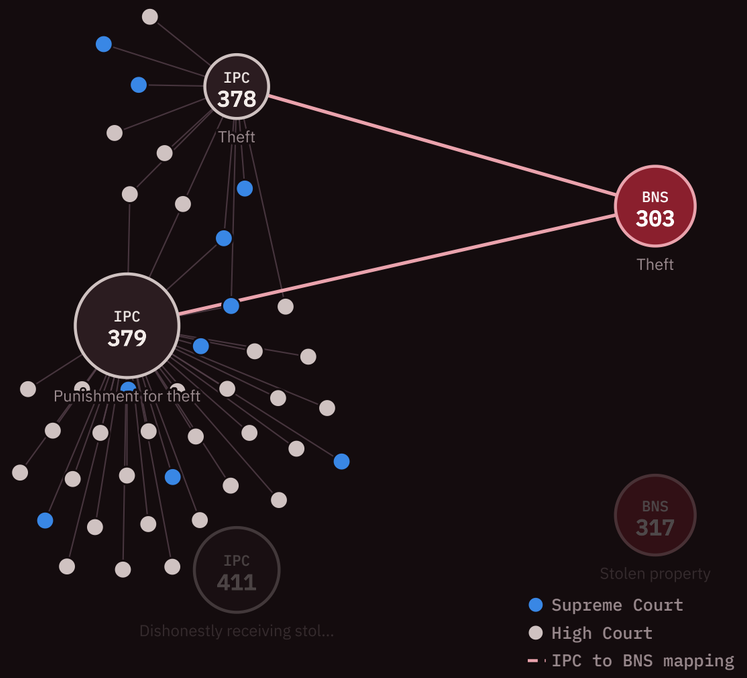
BNS 109	Attempt to murder	SUPPORTED
BNS 118(2)	Hurt by a dangerous weapon	SUPPORTED
BNS 105	Culpable homicide requires a death; the facts report grievous hurt, not death	DOUBTED
BNS 117	Grievous hurt	IN 118(2)

AGENT 4 · DEFENCE DRAFTER

3 counter-arguments ✓ each targets a charge

✓ every quote verbatim from the facts ✓ only supplied precedents cited

New law has almost no case law yet. We bridge it to the old.



BNS 303	0	judgments cite it directly
IPC 378	11	via the mapping
IPC 379	158	via the mapping

TOP 3 BY SIMILARITY TO THE FACTS

P1	Telangana High Court	0.686	Allowed
P2	Himachal Pradesh High Court	0.676	Allowed
P3	Delhi High Court	0.658	Dismissed

Agent 5 counts how appeals under each section were decided with plain SQL: IPC 379, 158 judgments: 76 allowed, 76 dismissed. Under 5 judgments, a figure is not shown.

The technology stack, and why each piece.

INTERFACE

FastAPI + web app

typed endpoints and live, stage-by-stage progress

ORCHESTRATION

LangGraph [12]

vs one big prompt: a retry edge, parallel agents, and a shared state checked step by step

INTELLIGENCE

qwen3.5:4b via Ollama · **BGE-M3** [8]

qwen3.5:4b: open-weight model with JSON-schema-constrained answers

BGE-M3 vs a legal-domain embedder: recall@3 0.533 vs 0.20, measured [10]

KNOWLEDGE

ChromaDB · **NetworkX** · **SQLite**

ChromaDB vs FAISS / Pinecone: persistent, with metadata filters

NetworkX: exact existence check · SQLite: exact counts

Python 3.12	**PyTorch** CUDA	**sentence-transformers** embedding
BGE-M3 1024-d vectors	**ChromaDB** vector store	**NetworkX** citation graph
LangGraph agents + state	**Ollama** qwen3.5:4b	**FastAPI** API + web app
SQLite outcomes	**PyMuPDF** PDF parsing	**pytest** 422 tests

07 · DEMO

Let's run it.

THE CASE

The complainant parked his motorcycle outside his house at night. In the morning it was gone. CCTV shows the accused pushing it away down the lane and riding off. The accused had no permission to take it and did not return it.

OFFENCE DATE 15 August 2024

WATCH FOR router picks BNS · 303 ranked first · verified in the graph · precedents via IPC 379

▶ LIVE

Switching to the browser

http://127.0.0.1:8000

Thank you.

Pramana: every section it shows exists in the code the offence date selects, and it says so when it doubts one.

RAG, not fine-tuning | BGE-M3 · ChromaDB · NetworkX | LangGraph multi-agent | qwen3.5:4b · FastAPI

Suvan Sharma · The Spartan Coders · Dept. of CSE, IKG PTU · Guide: Aditya Sharma

Questions welcome.

0

invented or wrong-code sections reached a report, across every evaluation we ran.

20 / 20
citation gate: real vs invented section numbers

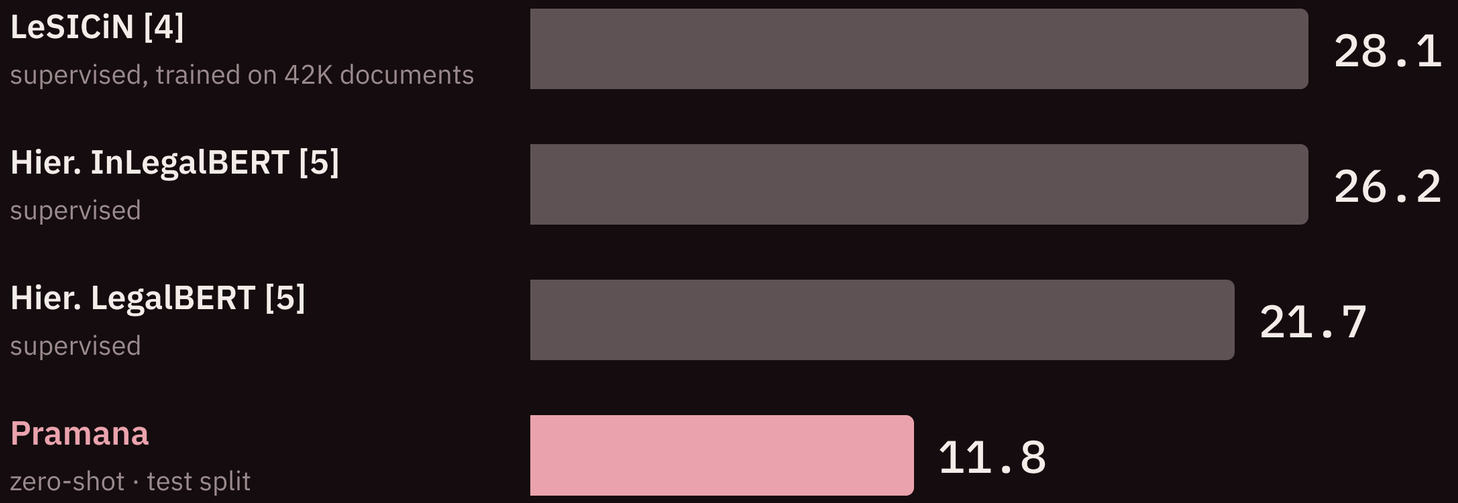
30 / 30
routing: the right code every time

889
citations checked across 263 saved runs; none unverified shown

6 / 6
cases recovered within 2 retries when the model cited from memory

Benchmarked on IL-TUR [5], with no task-specific training.

LEGAL STATUTE IDENTIFICATION · MACRO-F1 OVER 100 LABELS



100 test documents, run once after all development. Development split: 9.8. Micro precision / recall / F1: 40 / 21 / 27.

STRONG ON SERIOUS OFFENCES

302	Murder	0.78
376	Rape	0.63
34	Common intention	0.44

THE NEXT GAIN: WIDER RETRIEVAL

Minor companion offences such as simple hurt (323) or insult (504) rarely reach the candidate list: 37% of the right sections did.

What was hard, and what remains.

CHALLENGE → HOW WE HANDLED IT

No reliable digital BNS	→	parsed the official gazette PDF
Section numbers collide across Acts	→	code-scoped graph; CrPC citations filtered
The model invents sections	→	graph verification + retry loop
No BNS case law yet	→	IPC→BNS precedent bridge

LIMITATIONS

- Zero-shot section choice trails trained systems (11.8 vs 28.1)
- Minor companion offences often missed: 37% reach the candidate list
- IPC→BNS mapping not yet reviewed by practising lawyers
- Substantive criminal law only; English text; no scanned documents

The motorcycle case through every stage.

Router	**BNS**	15 Aug 2024 ≥ 1 Jul 2024
Retrieval	**14 candidates**	303(1) first: fused 0.0320, text rank 3, vote rank 2
Agent 2	**verified**	303 is a node of the BNS · 0 rejected
Agent 3	**3 precedents**	via IPC 379: Telangana 0.686 · HP 0.676 · Delhi 0.658
Agent 4	**1 argument**	dishonest intention not proved; quote verified verbatim
Guardrail	**pass**	closest judgment 0.686 ≥ 0.576
Agent 1	**proposes BNS 303**	from the candidate list
Agent 2b	**supported**	ingredients quoted from the facts, both passes
Agent 5	**IPC 379: 158**	76 allowed · 76 dismissed · 1 partly · 5 other · IPC 378: 8 / 3 · BNS 303: not shown
Total	**≈ 71 s**	end to end, from facts to report

One shared state, filled agent by agent.

`case_facts, offence_date`	input	from the lawyer
`era`	router	"ipc" or "bns"; the only code agent 2 verifies against
`retrieved`	retrieval	ranked candidates agent 1 reads
`candidate_sections`	agent 1	every round's proposals, tagged by round
`verified_sections, unconfirmed, retry_count`	agent 2	verified, rejected with the reason, retries used (≤ 2)
`applicability`	agent 2b	supported or doubted, with quotes and rules
`precedents`	agent 3	graph-linked judgments, IPC bridge for BNS
`analytics`	agent 5	SQL counts, not shown below 5 judgments
`defenses`	agent 4	counter-arguments, then the final check

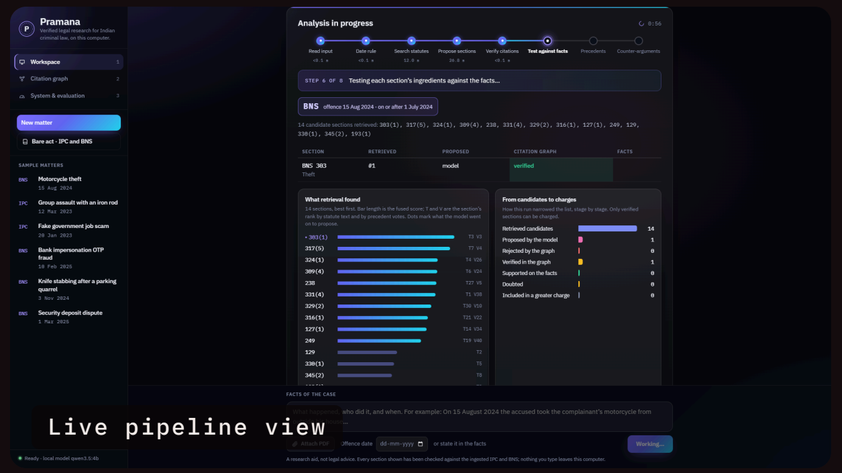
code · language model · graph lookup

How each result number was produced.

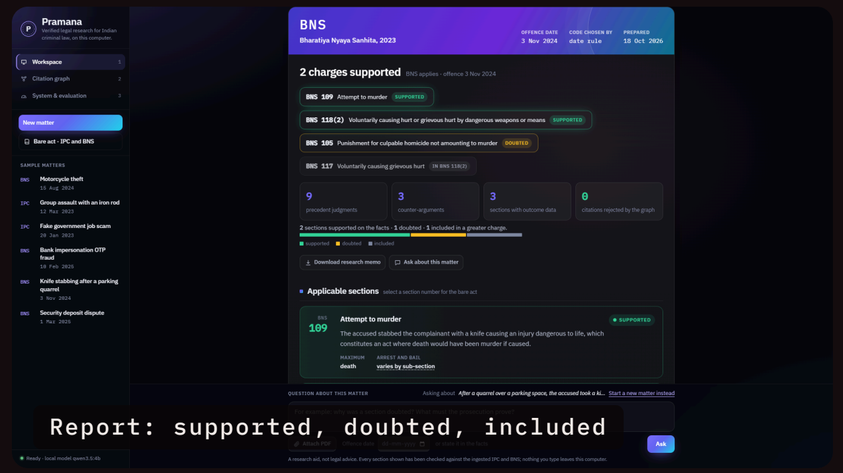
TEST	SET	HOW IT WAS RUN	RESULT
Citation gate	10 real + 10 invented numbers	verify_section on each, both codes	20 / 20
Routing	10 cases × 3 runs	code chosen and every verified section's code	30 / 30
Stress test	15 dev cases	retrieval withheld; model cites from memory	20 / 68 caught · 6 / 6 recovered
Real court orders	30 fresh Patna HC orders (CC-BY-4.0)	never read before scoring	13 / 30 right section
IL-TUR LSI	100 dev + 100 test documents	decisions on dev; test run once at the end	9.8 / 11.8 macro-F1
Embedder bake-off	15 hand-written cases	each searched only against its own code	0.533 vs 0.20 recall@3
Held-out sets	16 · 10 · 12 cases	written before each change, scored once	14/16 · 6/10 · 8/12

Hand-written sets were authored with AI assistance and are not lawyer-reviewed; the external orders and IL-TUR are independent.

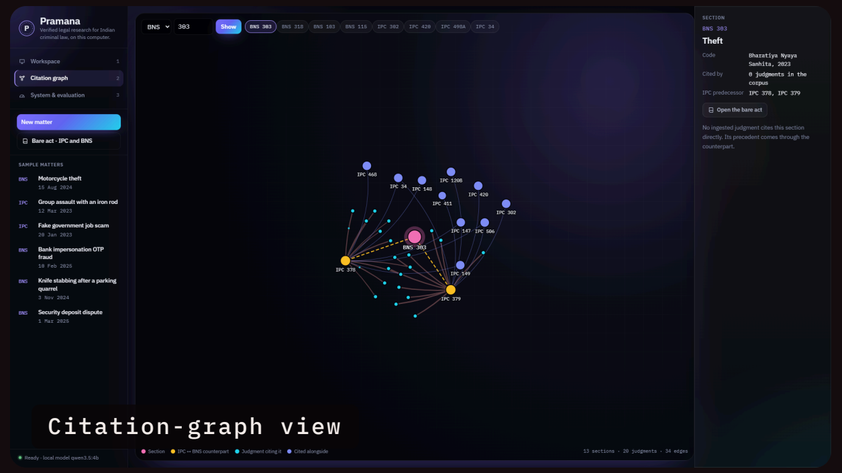
Verified and doubted are never confused.



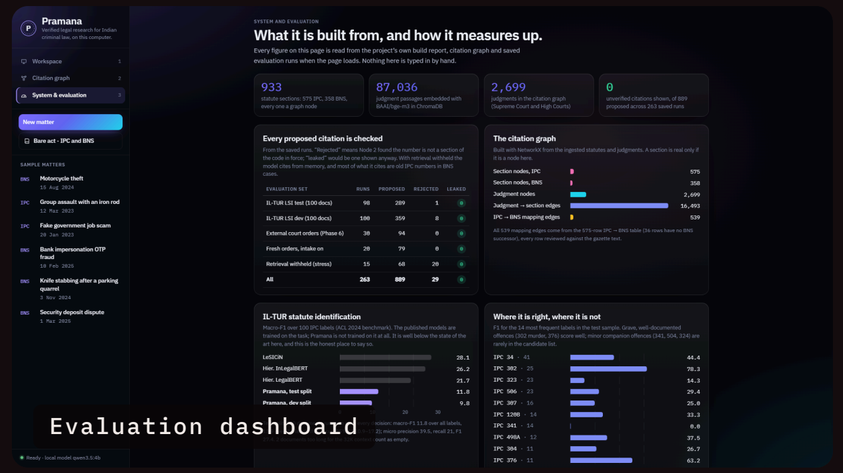
Live pipeline view



Report: supported, doubted, included



Citation-graph view



Evaluation dashboard

References.

[1] Government of India, “The Bharatiya Nyaya Sanhita, 2023 (Act No. 45 of 2023),” *Gazette of India*, Ministry of Law and Justice, 2023. [[VERIFY]]

[2] “The Indian Penal Code, 1860 (Act No. 45 of 1860),” Government of India. [[VERIFY]]

[3] P. Lewis *et al.*, “Retrieval-augmented generation for knowledge-intensive NLP tasks,” in *Proc. NeurIPS*, 2020. [[VERIFY]]

[4] S. Paul, P. Goyal and S. Ghosh, “LeSICiN: A heterogeneous graph-based approach for automatic legal statute identification from Indian legal documents,” in *Proc. AAAI*, 2022. [[VERIFY]]

[5] A. Joshi *et al.*, “IL-TUR: Benchmark for Indian legal text understanding and reasoning,” in *Proc. ACL*, 2024. [[VERIFY]]

[6] V. Magesh *et al.*, “Hallucination-free? Assessing the reliability of leading AI legal research tools,” arXiv:2405.20362, 2024. [[VERIFY]]

[7] M. Dahl, V. Magesh, M. Suzgun and D. E. Ho, “Large legal fictions: Profiling legal hallucinations in large language models,” *J. Legal Analysis*, vol. 16, 2024. [[VERIFY]]

[8] J. Chen *et al.*, “BGE M3-Embedding: Multi-lingual, multi-functionality, multi-granularity text embeddings through self-knowledge distillation,” arXiv:2402.03216, 2024. [[VERIFY]]

[9] G. V. Cormack, C. L. A. Clarke and S. Büttcher, “Reciprocal rank fusion outperforms Condorcet and individual rank learning methods,” in *Proc. SIGIR*, 2009. [[VERIFY]]

[10] D. Premasiri, T. Ranasinghe and R. Mitkov, “LLM-based embedders for prior case retrieval,” in *Proc. RANLP*, 2025. [[VERIFY]]

[11] S. K. Nigam *et al.*, “NyayaAnumana and INLegalLlama: The largest Indian legal judgment prediction dataset and specialized language model,” in *Proc. COLING*, 2025. [[VERIFY]]

[12] LangChain, “LangGraph,” github.com/langchain-ai/langgraph; Ollama, ollama.com; Chroma, trychroma.com. Software, 2026. [[VERIFY]]